

Assignment 2

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0.1 Instructions

This problem set is based on Lecture 7. It contains five conceptual questions on mathematical modeling of interest rates. The focus is not option valuation. The goal is to explain why short-rate models are used, when they are unnecessary, what drift and volatility assumptions mean, how arbitrage-free models differ from equilibrium models, and how volatility can be estimated from historical data.

For each question:

1. Explain the economic intuition before using formulas.
2. Be clear about what the model is trying to answer.
3. Distinguish today's observed yield curve from possible future yield curves.
4. Use concise reasoning. Do not over-mathematize.

1 Lecture 7: Interest-Rate Models

1.1 Question 1

A junior analyst says:

“Interest rates are uncertain, so we should use a short-rate model for every bond we price.”

The analyst is comparing the following securities:

Security	Description
A	5-year Treasury note with fixed annual coupons
B	5-year fixed-rate corporate bond with no embedded options
C	5-year callable agency bond
D	Interest-rate cap
E	Mortgage-backed security

Answer the following:

- Explain why Securities A and B usually do not require a stochastic short-rate model for time-0 valuation.
- State the basic valuation logic for a plain fixed-cash-flow bond using discount factors.
- Explain what a bootstrapped discount curve provides.
- Identify which securities in the table require future interest-rate dynamics and explain why.
- Correct the analyst’s statement in one or two sentences.

1.2 Question 2

A portfolio manager asks why Lecture 7 spends time modeling the short rate if today’s yield curve is already observable.

Answer the following:

- Explain the difference between a yield curve and a short-rate model.
- Explain what is meant by saying that a yield curve is a “snapshot.”
- Explain what is meant by saying that a short-rate model is a “scenario generator.”
- Explain why future discount curves $P(t, T)$ matter for securities whose cash flows depend on future rates.

- e. Give two examples of portfolio or risk-management questions that cannot be answered well using only today's yield curve.

1.3 Question 3

Consider the basic one-factor short-rate model:

$$dr_t = b(r_t, t)dt + \sigma(r_t, t)dz_t.$$

Answer the following:

- a. Explain the role of the drift term.
- b. Explain the role of the volatility term.
- c. Explain what mean reversion means and why it may be a reasonable assumption for interest rates.
- d. Compare the intuition behind constant volatility, level-proportional volatility, and square-root volatility.
- e. Explain one practical consequence of choosing a normal model that allows negative rates.

1.4 Question 4

A fixed-income team is deciding between an equilibrium model and an arbitrage-free model.

Answer the following:

- a. Explain the main purpose of an equilibrium interest-rate model.
- b. Explain the main purpose of an arbitrage-free interest-rate model.
- c. Explain why arbitrage-free models are usually preferred for pricing derivatives and bonds with embedded options.
- d. In the Hull-White model,

$$dr_t = [\theta(t) - ar_t]dt + \sigma dW_t,$$

explain why $\theta(t)$ is central to fitting today's curve.

e. Explain why a and σ are usually calibrated to interest-rate option prices rather than plain bond prices.

1.5 Question 5

An analyst wants to estimate historical short-rate volatility from weekly data. The analyst observes the following weekly changes in the short rate:

Week	Change in short rate
1	+4 bps
2	-2 bps
3	+6 bps
4	-5 bps
5	+3 bps
6	+1 bp
7	-4 bps
8	+5 bps

Answer the following:

- Explain why rate changes, rather than rate levels, are used to estimate volatility in this setting.
- Describe the steps for estimating weekly historical volatility from these observations.
- Compute the sample standard deviation of the weekly changes in basis points.
- Annualize the weekly volatility using $\sqrt{52}$.
- Explain two limitations of using this historical volatility estimate as an input to a short-rate model.